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Predicting Penguin Bill Length

A Bayesian Modeling Project

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1 Introduction

The goal of this work is to analyse the data collected by Gorman et al. (2014) on three species of *Pygoscelis* penguins inhabiting the Palmer Archipelago, Antarctica: Chinstrap, Gentoo, and Adelie. The central question is whether the bill length of these penguins can be predicted from other readily measured morphological and demographic characteristics. The bill length, or culmen, is one of the most important morphological indicators of Antarctic penguins (?): it relates to foraging behaviour and diet, and exhibits pronounced variation both between species and between sexes, which makes it a natural target for a structured statistical model.

We approach this prediction problem within a Bayesian framework, fitting all models in Stan (Stan Development Team, 2026) using Hamiltonian Monte Carlo. As predictors we use bill depth, a second continuous bill measurement, and sex, while species provides a natural grouping structure with three levels. Because the three species occupy clearly distinct regions of the bill depth–length space, a key modelling decision is how strongly to share information across species. We therefore fit and compare four models spanning the full pooling spectrum: a complete-pooling regression, two partial-pooling (hierarchical) models that let coefficients vary by species, and a no-pooling model with separate regressions per group. Hierarchical partial pooling is of particular interest here because the species groups are unbalanced and the smallest (Chinstrap) benefits from borrowing strength across groups.

The remainder of the report is organised as follows. Section 2 describes and explores the dataset. Section 3 introduces the four candidate models and justifies the likelihood and prior choices. Sections 4 and 5 present posterior predictive checks and leave-one-out cross-validation, respectively, and Section 6 assesses the sensitivity of the conclusions to the prior specification. Sections 7 and 8 discuss the findings and conclude.

2 Data

The dataset originates from Gorman et al. (2014), who collected morphological measurements of 333 penguins belonging to three *Pygoscelis* species: Adelie ($n = 146$), Chinstrap ($n = 68$), and Gentoo ($n = 119$). Each observation records the species and sex of the individual alongside two continuous bill measurements: bill depth (predictor, x) and bill length (outcome, y), both in millimetres. Bill depth and length were standardised prior to modelling, sex was coded as a binary indicator (female = 0, male = 1).

Table ?? summarises bill length by species and sex. Group means range from 37.3 mm (Adelie female) to 51.1 mm (Chinstrap male), and the within-group Pearson correlation ρ between bill depth and bill length is positive in five of the six groups, confirming the linear relationship motivating the use of bill depth as a continuous predictor.

Species	Sex	n	$\hat{\mu}$ (mm)	ρ
Adelie	Female	73	37.3	0.161
Adelie	Male	73	40.4	−0.038
Chinstrap	Female	34	46.6	0.256
Chinstrap	Male	34	51.1	0.446
Gentoo	Female	58	45.6	0.430
Gentoo	Male	61	49.5	0.307

Table 1: Descriptive statistics of bill length (mm) by species and sex. ρ is the Pearson correlation with bill depth within each group.

Figure 1 shows the distribution of bill length broken down by species and sex. Within each

species, the male and female distributions are clearly separated, with males consistently exhibiting longer bills. The magnitude of this sexual dimorphism appears roughly comparable across the three species. At the same time, the distributions differ substantially between species: Adelie penguins have the shortest bills (roughly 35–46 mm), while Chinstrap and Gentoo penguins are shifted upward into the 45–60 mm range. The within-group distributions are approximately unimodal and bell-shaped, which supports the use of a Gaussian likelihood. Two apparent outliers are visible: one female Chinstrap individual, and one male Gentoo, with unusually long bills; both lie at the tails of their respective group distributions but were retained in the analysis.

The scatter plot in Figure 2 reveals a positive linear relationship between bill depth and bill length within every species–sex combination. Crucially, the slopes and intercepts differ across species: Gentoo penguins occupy a distinct region of high depth and high length with a steep positive slope, whereas Adelie penguins span a wider range of bill depths but with a shallower slope and lower overall length. The vertical offset between female (solid line) and male (dashed line) trend lines is consistent in direction across all three species, yet the between-species differences in both intercept and slope are large enough to motivate fitting separate regression lines per group rather than assuming a single shared relationship. We therefore expect bill depth to be a meaningful positive predictor of bill length within each group, and that species and sex together will account for a large share of the between-individual variance in bill length.

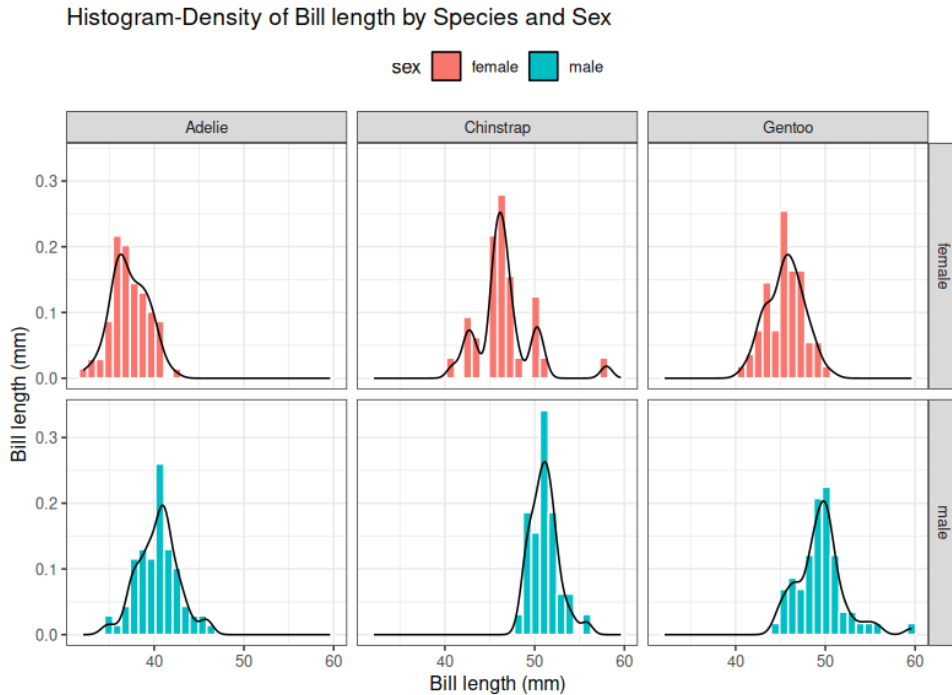


Figure 1: Histogram and kernel density of bill length (mm) by species (columns) and sex (rows).

3 Modelling

3.1 Observation model and variable selection

Bill length is a continuous morphological measurement recorded in millimetres. Within each species–sex combination the empirical distribution is approximately unimodal and symmetric (Figure 1), with no hard lower boundary in the observed range of roughly 32–60 mm. A Gaussian likelihood is therefore appropriate. A Student- t likelihood with heavier tails was considered as a robustness measure against the two apparent outliers identified in the data exploration (one female Chinstrap and one male Gentoo individual with an unusually long bill) however we don’t

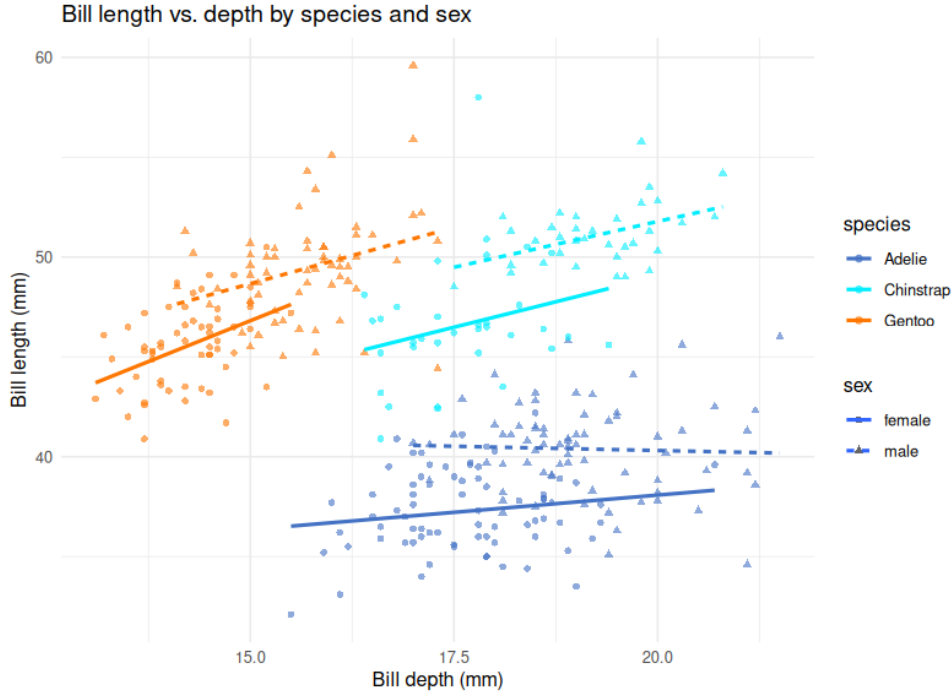


Figure 2: Bill length versus bill depth by species (colour) and sex (point shape / line type). Trend lines are ordinary least-squares fits within each group.

expect

All continuous predictors were standardised to zero mean and unit standard deviation prior to fitting. Standardisation places all regression coefficients on a common scale and stabilises sampler geometry, and it directly motivates the weakly informative $\text{Normal}(0, 1)$ priors described below.

Two predictors were selected on the basis of the data exploration and biological background. *Bill depth* (standardised, coded x) enters every model as a continuous covariate: the scatter plot (Figure 2) reveals a positive linear relationship between bill depth and bill length within every species–sex combination, making it the primary continuous driver of within-group variation. *Sex* (female = 0, male = 1) is included because a consistent male–female offset in bill length is visible across all three species, and omitting it would leave a large share of between-individual variance unexplained.

Species constitutes the grouping variable for all multi-group models. The three species occupy clearly distinct regions of the bill depth–length space with different intercepts and slopes (Figure 2), providing a strong empirical motivation for species-level structure. Group sizes are unequal and moderate (Adelie $n = 146$, Chinstrap $n = 68$, Gentoo $n = 119$), a setting in which partial pooling through a hierarchical prior can be particularly beneficial for the smaller group.

3.2 Models

Four models are fitted and compared, spanning the complete-pooling, partial-pooling, and no-pooling extremes. All models were implemented in Stan (Stan Development Team, 2026) and fitted using Hamiltonian Monte Carlo with four chains of 2 000 iterations each (1 000 warm-up), giving 4 000 post-warmup draws.

Model 1: Complete pooling The complete-pooling model treats every observation as exchangeable given the measured predictors. Species membership is absorbed by two fixed-effect dummy variables (d_{Chin} , d_{Gen} , with Adelie as baseline), and a single shared residual standard deviation σ is assumed.

$$\begin{aligned} \text{bill_length}_i &\sim \text{Normal}(\mu_i, \sigma) \\ \mu_i &= \alpha + \beta_{\text{depth}} x_i + \beta_{\text{sex}} \text{sex}_i + \beta_{\text{Chin}} d_{\text{Chin},i} + \beta_{\text{Gen}} d_{\text{Gen},i} \end{aligned} \quad (1)$$

Priors:

$$\begin{aligned} \alpha, \beta_{\text{depth}}, \beta_{\text{sex}}, \beta_{\text{Chin}}, \beta_{\text{Gen}} &\sim \text{Normal}(0, 1) \\ \sigma &\sim \text{Exponential}(1) \end{aligned} \quad (2)$$

Because both predictors are standardised, a coefficient of ± 1 on the unit-SD scale corresponds to a substantial but not extreme shift in bill length; $\text{Normal}(0, 1)$ therefore constitutes a weakly informative prior that rules out implausibly large effects without meaningfully constraining the posterior in the direction the data support. The $\text{Exponential}(1)$ prior on σ places most mass below roughly 3 mm, consistent with the within-group residual scatter seen in the data, while remaining proper and avoiding mass at zero.

The complete-pooling model provides a simple baseline. Its main limitation is that the species dummies only shift the intercept; different bill-depth slopes per species are not captured.

Model 2: Partial pooling, shared sex effect The first hierarchical model allows both the intercept and the bill-depth slope to vary by species, while the sex effect is shared (pooled) across species.

$$\begin{aligned} \text{bill_length}_i &\sim \text{Normal}(\mu_i, \sigma) \\ \mu_i &= \alpha_{s[i]} + \beta_{\text{depth},s[i]} x_i + \beta_{\text{sex}} \text{sex}_i \end{aligned} \quad (3)$$

Species-level priors (partial pooling):

$$\begin{aligned} \alpha_s &\sim \text{Normal}(\mu_\alpha, \sigma_\alpha) \quad s = 1, 2, 3 \\ \beta_{\text{depth},s} &\sim \text{Normal}(\mu_\beta, \sigma_\beta) \quad s = 1, 2, 3 \end{aligned} \quad (4)$$

Hyperpriors and remaining priors:

$$\begin{aligned} \mu_\alpha, \mu_\beta &\sim \text{Normal}(0, 1) \\ \sigma_\alpha, \sigma_\beta &\sim \text{Exponential}(1) \\ \beta_{\text{sex}} &\sim \text{Normal}(0, 1) \\ \sigma &\sim \text{Exponential}(1) \end{aligned} \quad (5)$$

The hyperprior means $\text{Normal}(0, 1)$ are weakly informative for the same reasons as in Model 1. The hyperprior standard deviations $\text{Exponential}(1)$ allow species to differ by a few standardised units in intercept and slope while keeping the prior proper; Gelman (2006) advocates half-Normal or Exponential hyperpriors for variance components in small hierarchies such as this one.

Pooling the sex effect is a deliberate simplification: the data exploration suggests that the male–female offset is roughly comparable across species, making a single shared coefficient a reasonable first approximation. This assumption is relaxed in Model 3.

Model 3: Partial pooling, species-specific sex effect Model 3 extends Model 2 by allowing the sex effect to vary by species as well, so that all three regression coefficients are species-specific and drawn from shared hyperpriors.

$$\begin{aligned} \text{bill_length}_i &\sim \text{Normal}(\mu_i, \sigma) \\ \mu_i &= \alpha_{s[i]} + \beta_{\text{depth},s[i]} x_i + \beta_{\text{sex},s[i]} \text{sex}_i \end{aligned} \quad (6)$$

Species-level priors:

$$\begin{aligned} \alpha_s &\sim \text{Normal}(\mu_\alpha, \sigma_\alpha) \\ \beta_{\text{depth},s} &\sim \text{Normal}(\mu_\beta, \sigma_\beta) \\ \beta_{\text{sex},s} &\sim \text{Normal}(\mu_{\text{sex}}, \sigma_{\text{sex}}) \end{aligned} \quad (7)$$

Hyperpriors:

$$\begin{aligned} \mu_\alpha, \mu_\beta, \mu_{\text{sex}} &\sim \text{Normal}(0, 1) \\ \sigma_\alpha, \sigma_\beta, \sigma_{\text{sex}} &\sim \text{Exponential}(1) \\ \sigma &\sim \text{Exponential}(1) \end{aligned} \quad (8)$$

The motivation for the added sex hierarchy is that, while the direction of sexual dimorphism is consistent across species, the histogram (Figure 1) suggests that the magnitude of the male–female separation may differ somewhat — most visibly, Chinstrap individuals show relatively tight male–female distributions compared with Adelie and Gentoo. Allowing a species-specific $\beta_{\text{sex},s}$ captures this variation while the shared hyperprior $\text{Normal}(\mu_{\text{sex}}, \sigma_{\text{sex}})$ prevents overfitting through partial pooling.

Model 4: No pooling, separate regressions The no-pooling model defines $G = 6$ independent groups from the full species $\times$ sex interaction (Adelie-female, Chinstrap-female, Gentoo-female, Adelie-male, Chinstrap-male, Gentoo-male). Each group receives its own intercept, bill-depth slope, and residual standard deviation, with no information shared across groups.

$$\begin{aligned} \text{bill_length}_i &\sim \text{Normal}(\mu_i, \sigma_{g[i]}) \\ \mu_i &= \alpha_{g[i]} + \beta_{g[i]} x_i \end{aligned} \quad (9)$$

Priors (independent per group):

$$\begin{aligned} \alpha_g &\sim \text{Normal}(0, 1) \quad g = 1, \dots, 6 \\ \beta_g &\sim \text{Normal}(0, 1) \\ \sigma_g &\sim \text{Exponential}(1) \end{aligned} \quad (10)$$

Sex is encoded structurally through the group index rather than as an explicit coefficient; male–female contrasts per species are recovered post-hoc from the posterior draws as $\alpha_{g_{\text{male}}} - \alpha_{g_{\text{female}}}$. Allowing a group-specific σ_g is a natural extension in this framework and accommodates the visual impression from Figure 1 that within-group spread differs somewhat across groups.

The no-pooling model is the most flexible of the four and serves as an upper bound on group-level heterogeneity. Its weakness is that it cannot borrow strength across groups: the Chinstrap group, being the smallest ($n_{\text{female}} = 34$, $n_{\text{male}} = 34$), may yield wider posterior intervals than the partial-pooling alternatives.

3.3 Prior justification

All continuous predictors were standardised before fitting, placing regression coefficients on a common scale where $\text{Normal}(0, 1)$ priors are weakly informative: they rule out implausibly large effects (beyond ± 3 SD units, i.e. ± 16 mm) while remaining essentially flat over the range the data support. The $\text{Exponential}(1)$ prior on all standard deviation parameters ($\sigma, \sigma_\alpha, \sigma_\beta, \sigma_{\text{sex}}$) concentrates mass near zero while accommodating the residual and between-species variation observed in the data.

To verify that the chosen priors are well-calibrated, we drew 1000 replicated datasets from the joint prior of each model and back-transformed the standardised predictions to millimetres. The resulting predictive distributions were centred near the observed range (32–60 mm) and assigned negligible probability to values outside $[0, 100]$ mm, confirming that the $\text{Normal}(0, 1)$ and $\text{Exponential}(1)$ priors are weakly informative and consistent with basic biological constraints on penguin bill length (Appendix A.4, Figures 8–10). The priors were retained unchanged.

4 Model checking

We evaluate all four models using posterior predictive checks (PPCs), comparing replicated datasets drawn from the posterior predictive distribution against the observed `bill_length` values.

The observed marginal distribution of `bill_length` is bimodal, with a larger left peak, a trough in the middle, and a right peak that itself shows a subtle secondary dip. The complete pooling model shifts predictive mass toward the right peak at the expense of the left, underestimating the left-hill density and overestimating the right; the secondary dip within the right peak is not recovered. Both hierarchical models centre their predictive envelopes more symmetrically across the two peaks, with the observed curve falling closer to the middle of the draw band. The separate model shows the widest spread of predictive draws in the central trough region, reflecting greater uncertainty from the smaller within-group sample sizes; the two main peaks are broadly recovered. The right tail decays more slowly than any model predicts, though the discrepancy is minor. The predictive density overlays for all four models are shown in Figure 3 below.

The posterior predictive distributions reproduce the observed mean, standard deviation, and minimum of `bill_length` well across all four models. The maximum is underestimated by all models: it corresponds to the two potential outliers identified in the data exploration, which lie beyond the range any model assigns meaningful predictive probability to under a Normal likelihood. Statistic checks are provided in Appendix A.3.

All four models converged without issue: $\hat{R} < 1.01$ throughout and traceplots show well-mixed chains. The fully hierarchical model (v3) shows somewhat lower effective sample sizes, consistent with its higher-dimensional parameter space and additional pooling structure on β_{sex} . Detailed $\hat{R}$ and ESS values for all models are provided in Table 5 in the Appendix. Overall, the hierarchical and separate models provide a more faithful representation of the data than complete pooling, motivating the formal comparison in the following chapter.

5 Model comparison

For model comparison we examined the predictive performance of each model using leave-one-out cross-validation, checked its validity via Pareto- k values for each observation, and assessed tail performance via estimation of minima and maxima. The overall posterior predictive check

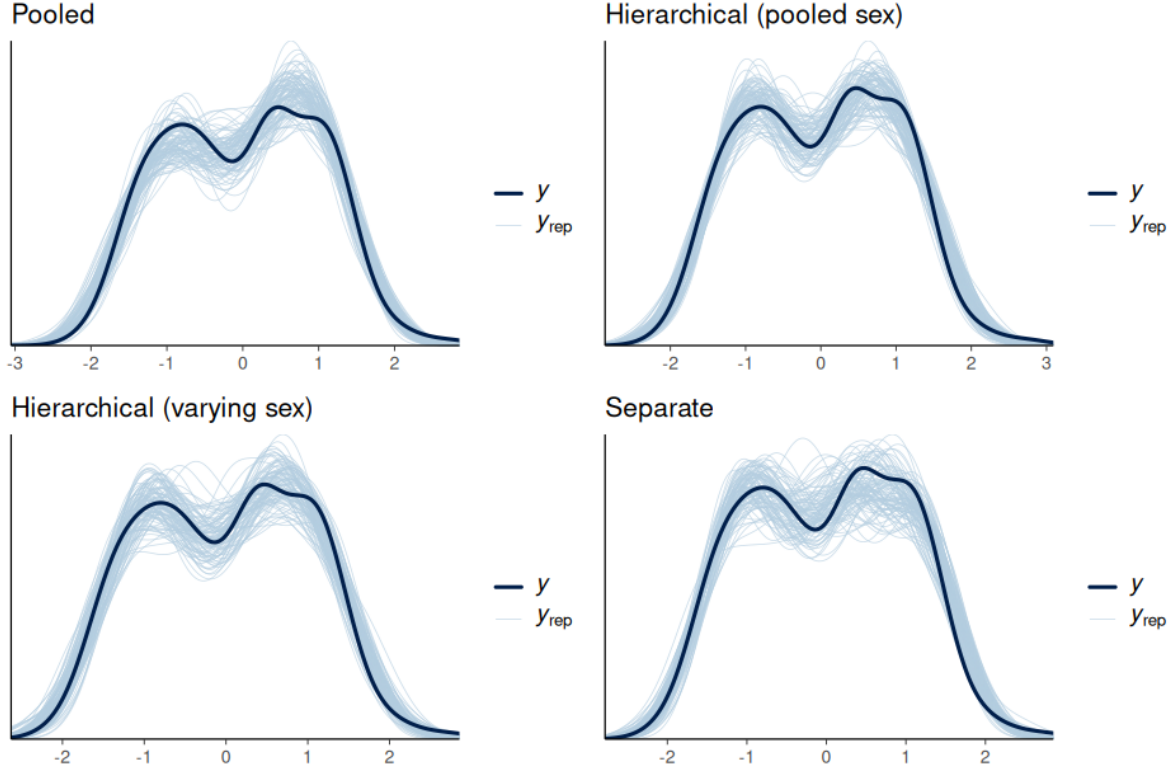


Figure 3: Predictive density overlays for all four models. The dark line is the observed `bill_length` distribution; light lines are 100 draws from the posterior predictive distribution.

plots, in which multiple draws from the posterior are plotted against the sample distribution, are also considered.

The posterior predictive plots (Figure 3) show that the pooled model overestimates the Chinstrap and Gentoo region while underestimating the Adelie part of the joint distribution. The hierarchical models correct these shortcomings, while the pooled sex effect variant captures the right tail slightly better than the other models. The separate model shows higher variability in the modes of the joint distribution and displays a tendency to underestimate the Chinstrap and Gentoo area.

Model comparison using leave-one-out cross-validation (Table 2) indicates that the separate model achieved the highest expected log predictive density ($\text{ELPD}_{\text{LOO}} = -178.3$), suggesting the strongest out-of-sample predictive performance. The hierarchical models exhibited only slightly lower predictive performance ($\Delta\text{ELPD} = -3.3$ and -3.8 relative to the best model), with standard errors exceeding the observed differences (Table 2). This suggests that the predictive advantage of the fully separate model is not significant. The complete pooling model performed worst, with an ELPD difference of -7.8 . All models exhibited satisfactory Pareto- k diagnostics ($k < 0.7$ for all observations), indicating reliable PSIS-LOO estimates.

Model 4 achieved the highest expected log predictive density ($\text{ELPD}_{\text{LOO}} = -178$), indicating the best estimated out-of-sample predictive performance among the candidate models. However, the improvement over the hierarchical models (approximately 4 ELPD units) is small relative to the associated standard errors (16–19), suggesting that the predictive advantage is uncertain. The separate model also exhibits substantially greater effective complexity ($p_{\text{LOO}} = 21.3$) compared with the hierarchical models ($p_{\text{LOO}} \approx 10$), since it effectively fits six individual linear regression models for each species–sex combination. The hierarchical models provide a compromise between

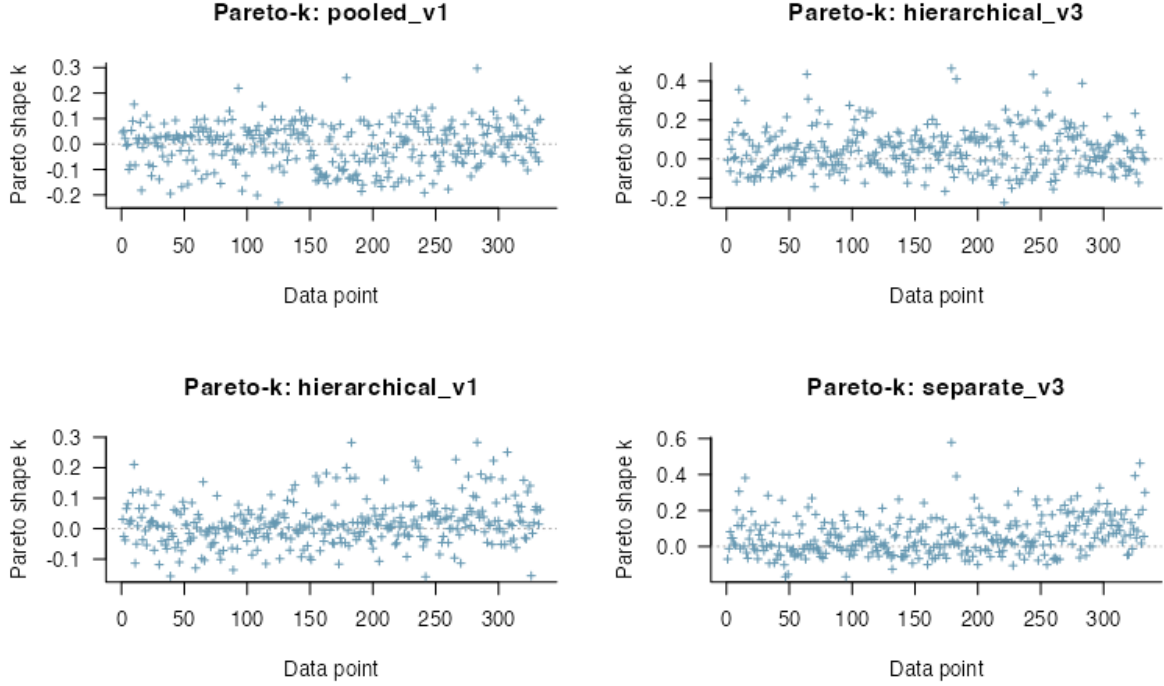


Figure 4: Pareto- k values for Model 1–4.

predictive accuracy and model complexity, while the pooled model performs worst in terms of out-of-sample predictive performance.

Table 2: Model comparison based on LOO cross-validation (sorted by ELPD; higher values are better).

Model	ELPD _{LOO}	SE(ELPD)	p_{LOO}	LOOIC
separate_v3	−178	16.8	21.3	357
hierarchical_v1	−182	19.1	10.1	363
hierarchical_v3	−182	19.2	10.9	364
pooled_v1	−186	18.9	7.02	372

6 Sensitivity analysis

For robustness and sensitivity checks we refit all four models with a Student- t distribution with 3 degrees of freedom and a uniform distribution on the interval $(-100, 100)$.

As a more robust alternative to the chosen normal priors, we tested the models with a Student- t distribution with three degrees of freedom, centred at zero with a dispersion parameter of one. The overall posterior predictive checks are very similar to the base models. Nevertheless all four new models showed slightly worse performance in the right tail of the distribution, underestimating the maximum even more than the base models. In terms of convergence, the fully separate model had one observation with a Pareto- k between 0.7 and 1.0; all other models showed no observations above the generally accepted threshold. The LOO statistics yield the same results as with normal priors.

As a second alternative we tested uniform priors across all parameters, with bounds $(-100, 100)$ for intercepts and slopes and $(0, 100)$ for the standard deviation. Given the standardised data,

this is such a wide prior that we expect the posterior to be dominated entirely by the likelihood. The posterior predictive checks again yield little difference compared to the base models. Only the separate model shows better performance in the right tail. Examining the minima and maxima statistics directly, the pooled and hierarchical models underestimate the maximum more than the base model while retaining a good fit for the minimum. The min–max plot for the fully separate model shows, consistent with the posterior predictive check, a better estimate for the maximum, along with a negligible shift to the left in minimum estimation. The reliability analysis shows good Pareto- k values for all models, with one observation between 0.7 and 1.0. The LOO scores are also essentially unchanged under this prior regime.

The models show very little change in behaviour under alternative prior specifications. The underestimation of the maximum is a common weakness across all three prior choices and is attributed to the outlier structure of the data rather than the prior. The one observation with elevated Pareto- k was the same female Chinstrap penguin with a bill length of 58 mm in both cases, approximately 3.7 standard deviations from the female Chinstrap mean. Since the separate model fits six independent linear regressions for each species–sex combination, this is not unexpected and explains the slight improvement in maximum estimation under a uniform prior.

Model	ELPD _{LOO}	SE(ELPD)	p_{LOO}	LOOIC
separate	−178	16.7	21.0	356
hierarchical (var. sex)	−182	19.2	10.9	364
hierarchical (pooled sex)	−182	19.2	10.9	364
pooled	−186	18.9	7.14	372

Table 3: LOO comparison for Student- t priors

Model	ELPD _{LOO}	SE(ELPD)	p_{LOO}	LOOIC
separate	−178	16.5	21.3	357
hierarchical (var. sex)	−182	19.2	11.2	364
hierarchical (pooled sex)	−182	19.2	11.2	364
pooled	−186	18.9	7.15	372

Table 4: LOO comparison for uniform priors

7 Discussion

Findings The four models span the complete-pooling, partial-pooling, and no-pooling extremes, and the comparison yields a coherent picture rather than a single clear winner. The no-pooling separate model achieved the best point estimates of out-of-sample predictive performance (highest ELPD_{LOO}, lowest LOOIC), but it also carried by far the highest effective complexity ($p_{\text{LOO}} = 21.3$) and the differences in ELPD relative to the hierarchical models (3–4 units) were well within their standard errors (16–19). On purely predictive grounds, therefore, no model is significantly preferable to the others, and the apparent edge of the separate model cannot be distinguished from sampling noise. Complete pooling, which represents species only as intercept shifts and forces a single shared depth slope, ranked worst on point estimates, but even this gap (roughly 8 ELPD units) is small relative to the standard errors.

The two hierarchical models performed essentially identically, even though the pooled-sex variant (Model 2) is the simpler of the two. That the species-specific sex effect in Model 3 buys no measurable improvement suggests that the degree of sexual dimorphism is roughly constant

across species, so a single shared male–female offset is an adequate description. This is consistent with the data exploration, where the male–female separation appeared comparable across the three species.

Both the posterior predictive checks and the comparison highlight the right tail of the bill-length distribution as the common weakness: every model underestimates the maximum, driven by the male Gentoo and female Chinstrap extremes that lie beyond the range a Normal likelihood assigns meaningful mass. This points to the outliers, rather than the pooling structure, as the limiting factor for tail fit.

Limitations and potential improvements The clearest limitation is the treatment of the tails. A Student- t likelihood (e.g. with three degrees of freedom) is worth exploring as a more principled way to accommodate the extreme observations; we used t -distributed *priors* as a robustness check, but a heavier-tailed *likelihood* would target the min/max estimates directly and may yield better tail coverage than the Normal model.

A second consideration concerns the smallest group. The Chinstrap species has the fewest observations ($n = 68$, only 34 per sex), and the no-pooling separate model cannot borrow strength across groups in this regime. Indeed the single persistently high Pareto- k observation under the alternative priors was a female Chinstrap individual at the tail of its group. The hierarchical models are better suited to such unbalanced, small-cell data because they share information across species through the hyperprior, which is a substantive argument in their favour beyond the raw predictive scores.

Finally, the analysis is limited to two morphological predictors. Incorporating additional measurements such as body mass or flipper length, or combining the species and sex hierarchies more richly, could explain further between-individual variance. A natural extension would be to ask the inverse predictive question: whether species and sex can be classified from bill measurements alone.

8 Conclusion

We set out to predict the bill length of Adelie, Chinstrap, and Gentoo penguins from bill depth and sex, fitting four Bayesian linear models in Stan that ranged from complete pooling to fully separate group-wise regressions. Bill depth is a positive predictor of bill length within every species–sex group, males have consistently longer bills than females, and the species differ markedly in both intercept and slope. Complete pooling captures species only as intercept shifts through fixed dummy variables, whereas the hierarchical and separate models let the depth slope (and, for the separate model, the residual scale) vary by species; the latter fit the data better, though by a margin that is not significant relative to the LOO standard errors.

Among these, leave-one-out cross-validation does not single out one model: the separate model has the best point estimates but the differences from the hierarchical models are not significant and come at a substantial cost in complexity. We therefore favour the hierarchical model with a pooled sex effect, which matches the predictive performance of the more complex alternatives while sharing information across the unbalanced species groups. The main shared weakness is the underestimation of the longest bills, which a heavier-tailed likelihood could address. The sensitivity analysis showed that replacing the weakly informative priors with Student- t or very wide uniform priors left the posterior summaries and LOO scores essentially unchanged, indicating that the conclusions are driven by the data and are robust to the prior specification.

9 AI Disclosure

Portions of this project were developed with the assistance of AI tools. Specifically, AI was used to support code development, data visualization and the drafting or refinement of written content. All AI-generated outputs were reviewed, edited and verified by the authors, who take full responsibility for the accuracy and integrity of the final work.

References

- Gelman, A. (2006). Prior distributions for variance parameters in hierarchical models (comment on article by browne and draper). *Bayesian Analysis*, 1.
- Gorman, K. B., Williams, T. D., and Fraser, W. R. (2014). Ecological sexual dimorphism and environmental variability within a community of antarctic penguins (genus *pygoscelis*). *PLOS ONE*, 9(3):1–14.
- Stan Development Team (2026). Stan modeling language users guide and reference manual. <https://mc-stan.org>.

A Appendix

A.1 Stan model code

A.2 Full convergence diagnostics

Model	Parameter	Mean	SD	5%	50%	95%	n_{eff}	$\hat{R}$
Pooled	α	-1.315	0.044	-1.387	-1.315	-1.241	3172	1.00
	β_{depth}	0.180	0.054	0.090	0.180	0.271	2595	1.00
	β_{sex}	0.535	0.063	0.433	0.534	0.638	2963	1.00
	$\beta_{\text{species}[1]}$	1.811	0.062	1.707	1.811	1.912	3530	1.00
	$\beta_{\text{species}[2]}$	1.890	0.106	1.714	1.892	2.063	2548	1.00
	σ	0.419	0.017	0.393	0.419	0.448	3195	1.00
Hierarchical v1	μ_{α}	-0.010	0.563	-0.932	-0.001	0.897	2986	1.00
	$\mu_{\beta_{\text{depth}}}$	0.252	0.227	-0.095	0.251	0.601	1621	1.00
	σ_{α}	1.222	0.585	0.597	1.066	2.338	2720	1.00
	σ_{β}	0.346	0.295	0.096	0.259	0.863	1629	1.00
	$\alpha[1]$	-1.235	0.050	-1.318	-1.235	-1.152	3257	1.00
	$\alpha[2]$	0.407	0.075	0.282	0.408	0.532	2509	1.00
	$\alpha[3]$	0.786	0.122	0.592	0.784	0.986	1874	1.00
	$\beta_{\text{depth}[1]}$	0.076	0.064	-0.033	0.076	0.179	2781	1.00
	$\beta_{\text{depth}[2]}$	0.360	0.093	0.210	0.358	0.515	2476	1.00
	$\beta_{\text{depth}[3]}$	0.351	0.088	0.208	0.351	0.495	2042	1.00
	β_{sex}	0.493	0.062	0.392	0.494	0.594	2109	1.00
	σ	0.412	0.017	0.385	0.411	0.440	4202	1.00
Hierarchical v3	μ_{α}	0.006	0.564	-0.932	0.017	0.918	1952	1.00
	$\mu_{\beta_{\text{depth}}}$	0.247	0.237	-0.135	0.244	0.596	1372	1.01
	μ_{sex}	0.478	0.169	0.223	0.489	0.704	1299	1.00
	σ_{α}	1.223	0.556	0.623	1.098	2.300	2226	1.00
	σ_{β}	0.368	0.315	0.101	0.272	0.936	1112	1.01
	σ_{sex}	0.195	0.241	0.021	0.120	0.629	1004	1.01
	$\alpha[1]$	-1.242	0.052	-1.324	-1.242	-1.156	1930	1.00
	$\alpha[2]$	0.402	0.074	0.283	0.401	0.521	1509	1.00
	$\alpha[3]$	0.847	0.161	0.599	0.837	1.124	1222	1.00
	$\beta_{\text{depth}[1]}$	0.067	0.067	-0.045	0.068	0.176	1780	1.00
	$\beta_{\text{depth}[2]}$	0.346	0.107	0.171	0.347	0.526	1506	1.00
	$\beta_{\text{depth}[3]}$	0.384	0.105	0.219	0.379	0.566	1226	1.00
	$\beta_{\text{sex}[1]}$	0.515	0.076	0.395	0.512	0.643	2355	1.00
	$\beta_{\text{sex}[2]}$	0.519	0.110	0.345	0.516	0.707	1821	1.00
	$\beta_{\text{sex}[3]}$	0.448	0.100	0.275	0.457	0.597	820	1.01
	σ	0.412	0.016	0.386	0.412	0.439	3806	1.00
Separate	$\alpha[1]$	-1.258	0.049	-1.337	-1.258	-1.176	4757	1.00
	$\alpha[2]$	0.392	0.113	0.209	0.394	0.577	3812	1.00
	$\alpha[3]$	1.067	0.248	0.667	1.068	1.470	2766	1.00
	$\alpha[4]$	-0.623	0.105	-0.800	-0.622	-0.451	2919	1.00
	$\alpha[5]$	0.935	0.132	0.722	0.936	1.154	3083	1.00
	$\alpha[6]$	1.265	0.136	1.040	1.266	1.493	2737	1.00
	$\beta[1]$	0.122	0.091	-0.026	0.123	0.272	4889	1.00
	$\beta[2]$	0.347	0.250	-0.064	0.346	0.761	3746	1.00
	$\beta[3]$	0.527	0.164	0.256	0.525	0.792	2805	1.00

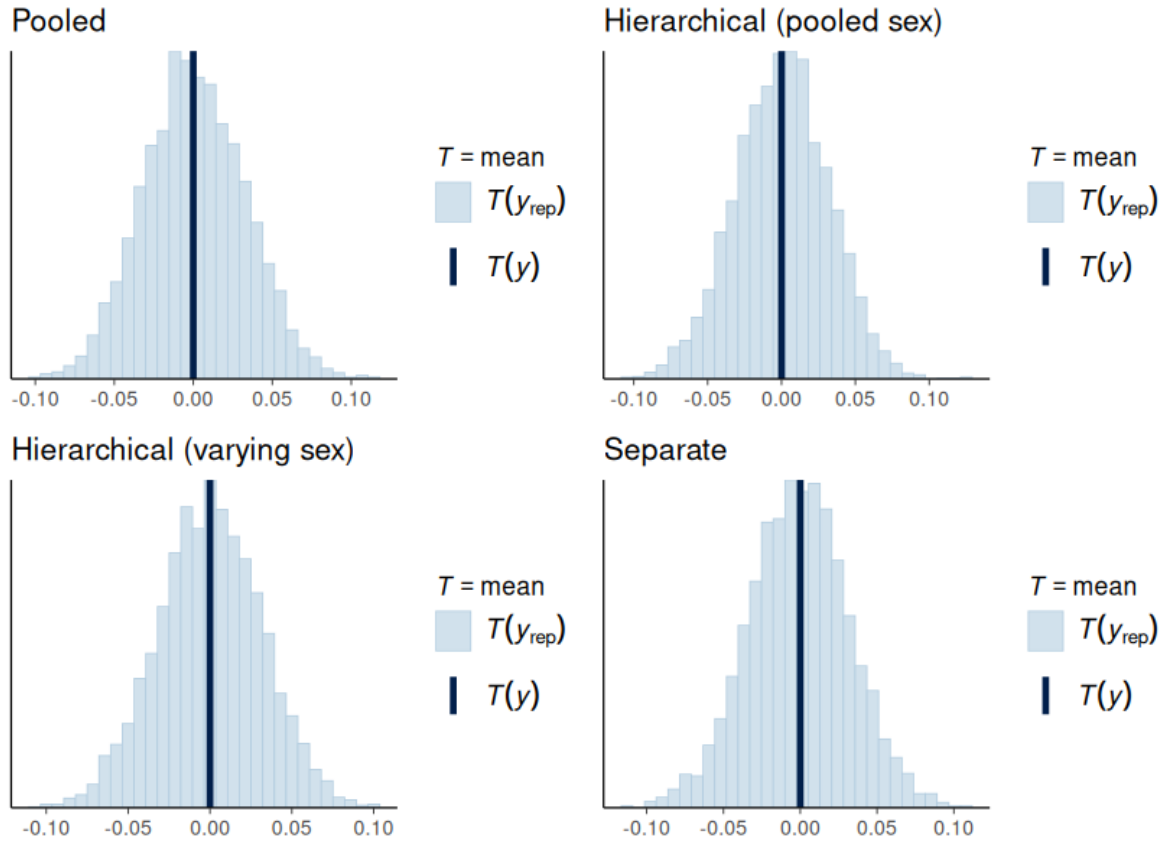
Table 5 (continued)

Model	Parameter	Mean	SD	5%	50%	95%	n_{eff}	$\hat{R}$
	$\beta[4]$	-0.035	0.095	-0.188	-0.037	0.124	2978	1.00
	$\beta[5]$	0.341	0.117	0.152	0.340	0.533	3043	1.00
	$\beta[6]$	0.364	0.166	0.093	0.366	0.635	2533	1.00
	$\sigma[1]$	0.374	0.032	0.326	0.372	0.432	5028	1.00
	$\sigma[2]$	0.574	0.075	0.467	0.565	0.708	3605	1.00
	$\sigma[3]$	0.348	0.035	0.296	0.345	0.410	4714	1.00
	$\sigma[4]$	0.426	0.038	0.368	0.422	0.493	4493	1.00
	$\sigma[5]$	0.269	0.036	0.217	0.265	0.336	4259	1.00
	$\sigma[6]$	0.486	0.046	0.416	0.482	0.566	4880	1.00

Group indices for the Separate model: 1 = Adelie female, 2 = Chinstrap female, 3 = Gentoo female, 4 = Adelie male, 5 = Chinstrap male, 6 = Gentoo male.

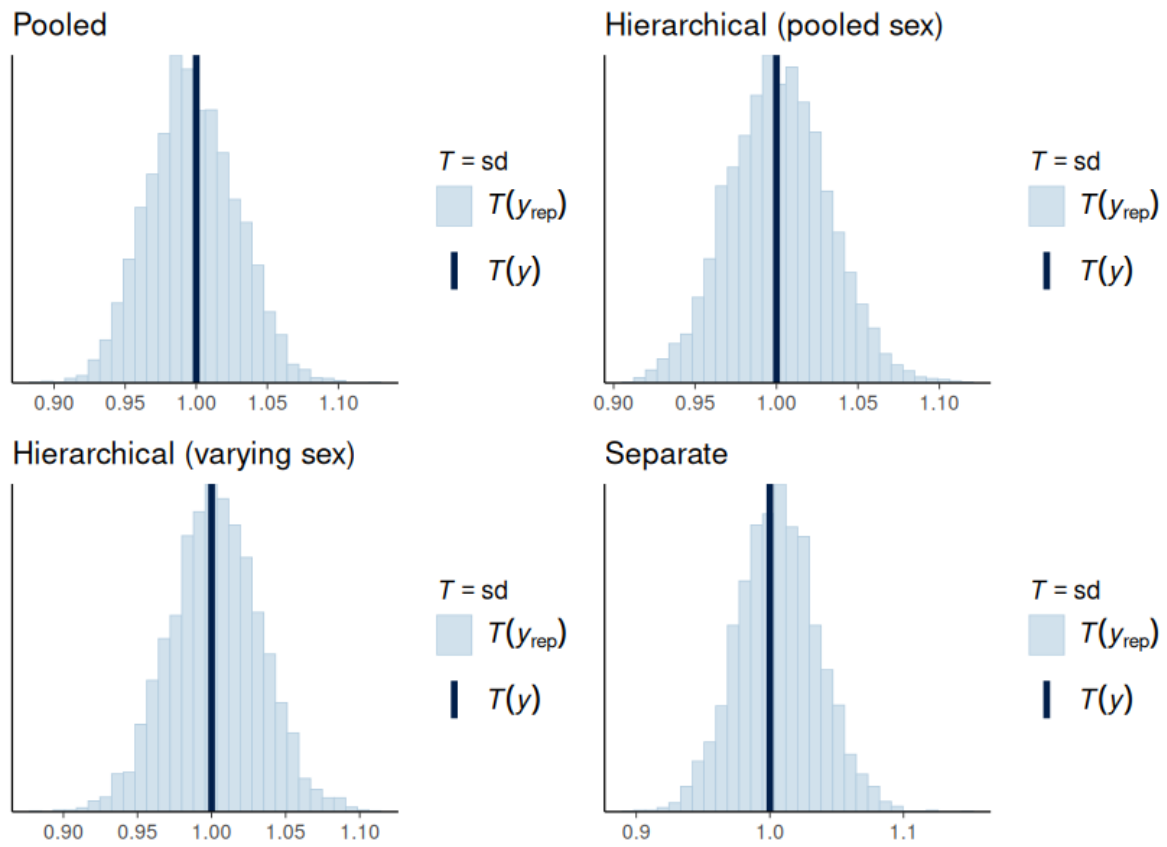
Table 5: Posterior summaries and convergence diagnostics across all four model parameters.

A.3 Posterior predictive checks



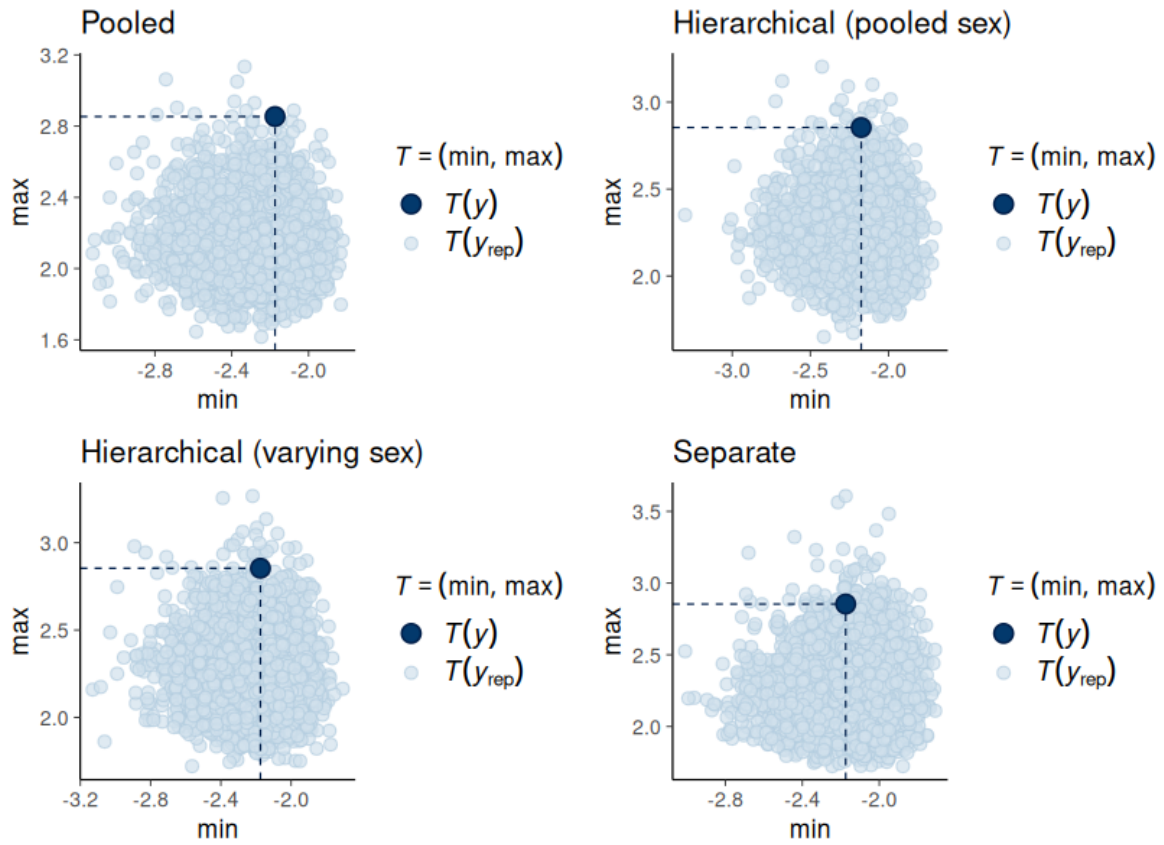
Posterior predictive checks for the mean across all four models. The dark line is the observed mean of `bill_length`; the histogram shows the distribution of means from posterior predictive draws.

Figure 5: Posterior predictive check on the mean of `bill_length` for all four models. The vertical line marks the observed statistic.



Posterior predictive checks for the standard deviation across all four models. The dark line is the observed SD of `bill_length`; the histogram shows the distribution of SDs from posterior predictive draws.

Figure 6: Posterior predictive check on the standard deviation of `bill_length` for all four models.



Posterior predictive checks for the min and max across all four models. The dark point is the observed (min, max) of bill_length; the scatter shows (min, max) pairs from posterior predictive draws.

Figure 7: Posterior predictive check on the minimum and maximum of `bill_length` for all four models. The observed maximum falls in the upper tail of the predictive distribution across all models.

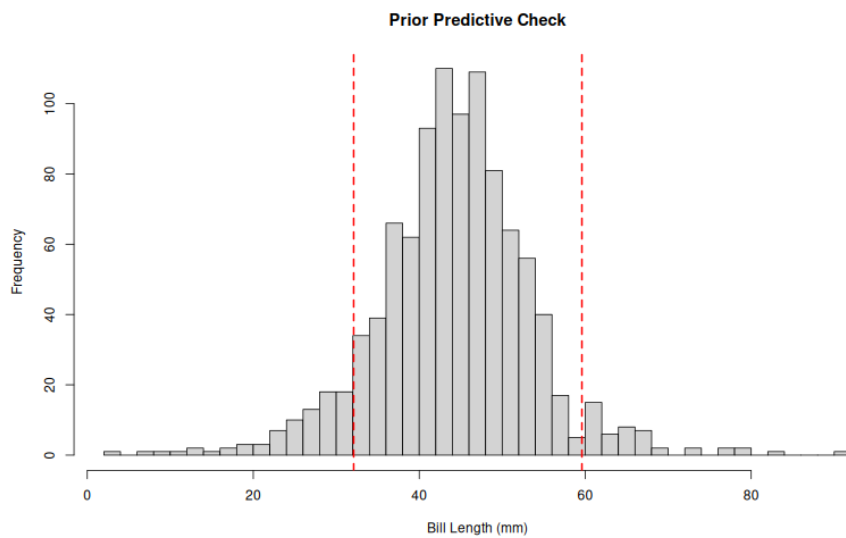


Figure 8: Marginal prior predictive distribution of bill length (mm) for the complete-pooling model. Red dashed lines mark the observed range (32–60 mm).

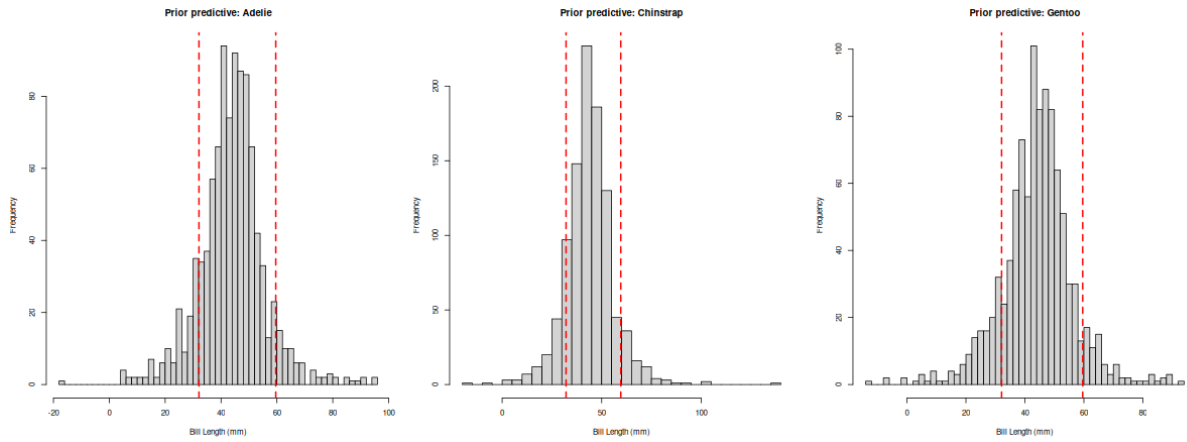


Figure 9: Prior predictive distributions per species under the hierarchical prior, evaluated at mean bill depth and for female individuals (sex = 0). Red dashed lines mark the observed range.

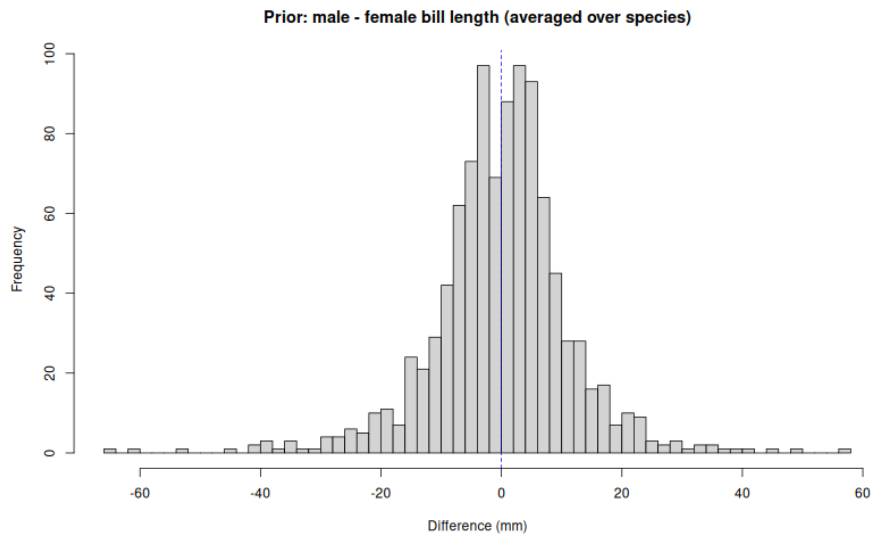


Figure 10: Prior distribution of the male–female bill length contrast (mm), averaged across species. The blue dashed line marks zero.

A.4 Prior predictive checks

A.5 Sensitivity Analysis

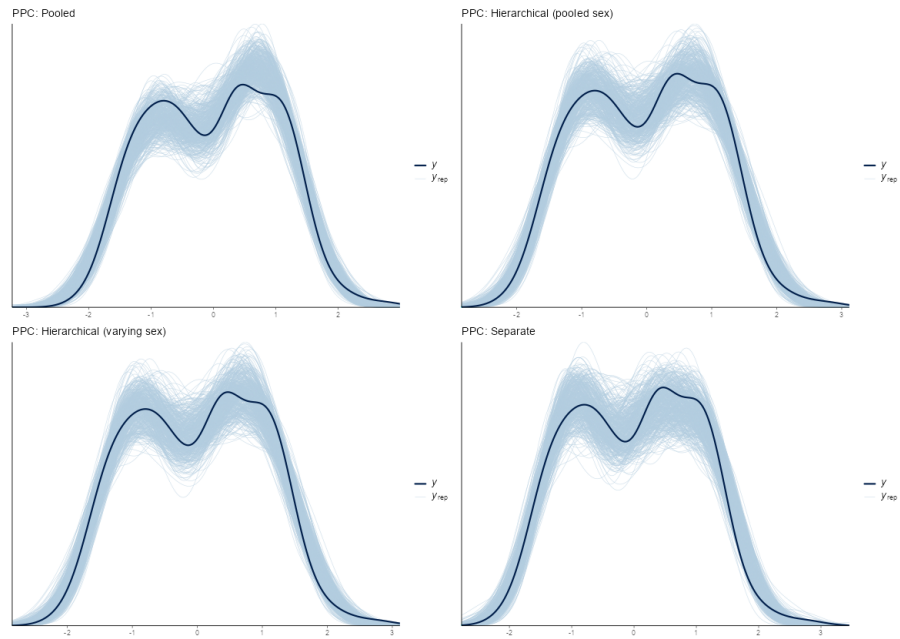


Figure 11: Predictive density overlays for all four models with student-t priors. The dark line is the observed `bill_length` distribution, light lines are 100 draws from the posterior predictive distribution.

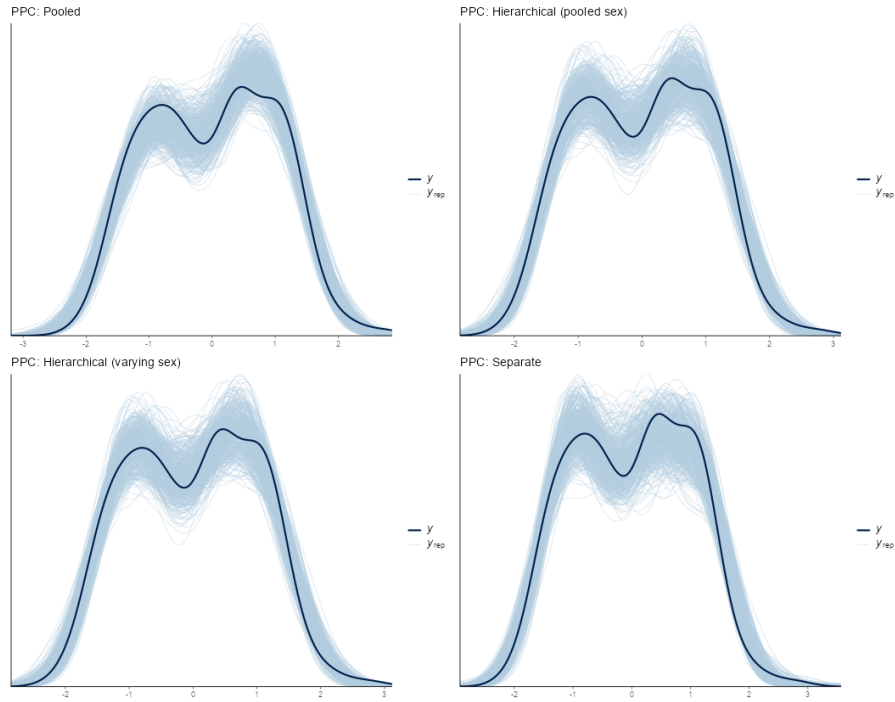


Figure 12: Predictive density overlays for all four models with uniform priors. The dark line is the observed `bill_length` distribution; light lines are 100 draws from the posterior predictive distribution.

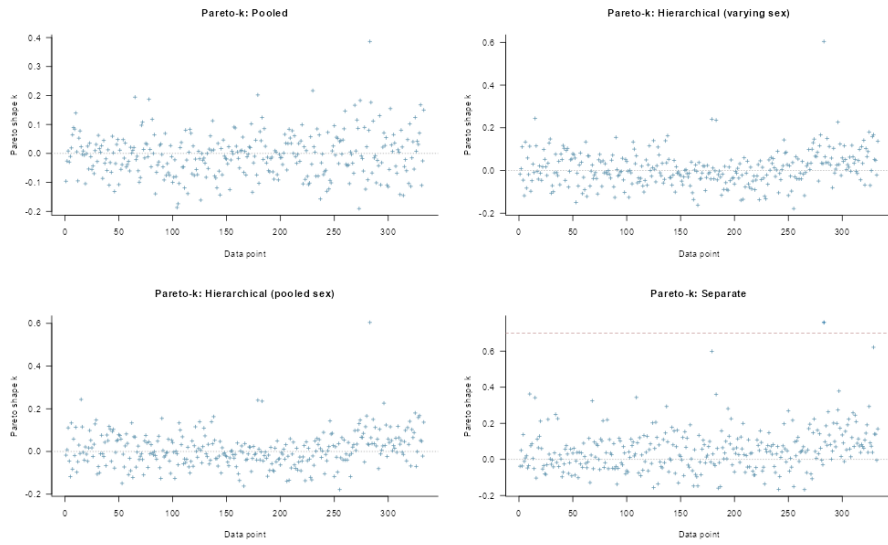


Figure 13: Pareto-k values for Model 1—4 with student-t priors.

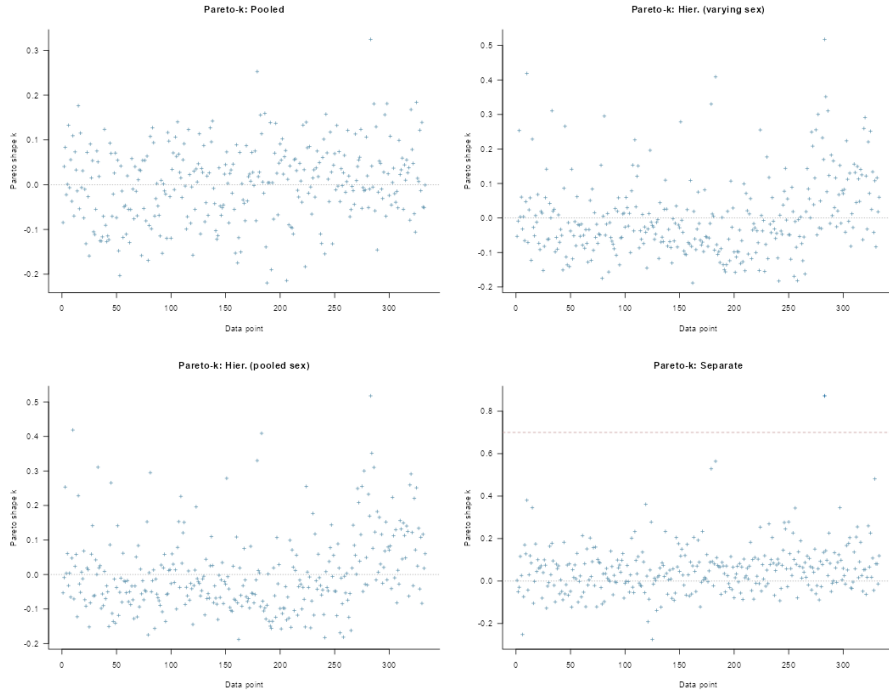


Figure 14: Pareto-k values for Model 1—4 with uniform priors.

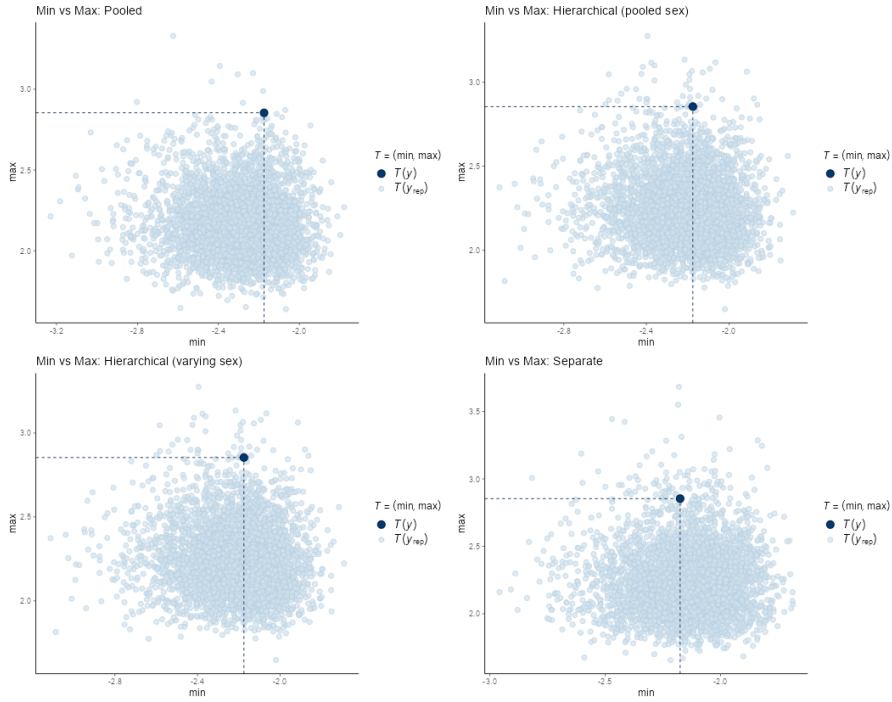


Figure 15: Posterior predictive check on the minimum and maximum of `bill_length` for all four models with student-t priors.

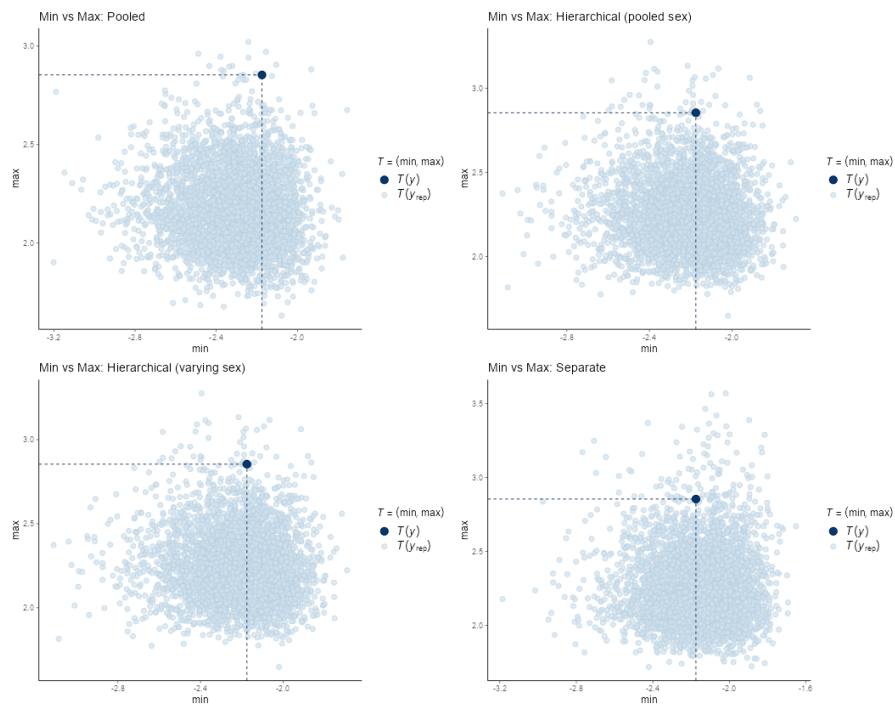


Figure 16: Posterior predictive check on the minimum and maximum of `bill_length` for all four models with uniform priors.